# AIRLINE ANALYSIS REPORT

Tableau Dashboard — Delayed Flights Dataset

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## Project Overview

This report presents a comprehensive visual analysis of airline flight data built using Tableau. The dashboard was created to provide insights into flight volumes, delay patterns, route distributions, and key performance metrics across multiple U.S. airline carriers. The underlying dataset used is the **DelayedFlights** dataset, which captures flight-level records including departure/arrival delays, carrier codes, origin airports, and elapsed time.

The analysis is structured across two dashboards and several individual worksheets, each designed to answer specific business questions about airline performance.

### Key Metrics at a Glance

| Metric | Value | Description |
|---|---|---|
| **Total Flights KPI** | 19,36,758 | Total number of flights recorded in the dataset |
| **Total Flights Delay** | 83,639,236 | Aggregate delay minutes across all flights |
| **Average Delay** | 42.20 mins | Mean delay per flight across all carriers |
| **No. of Carriers** | 20 | Unique airline carriers in the dataset |
| **No. of Origin Airports** | 10+ | Key origin airports tracked in the analysis |

# Dashboard 2 — Airline Analysis Overview

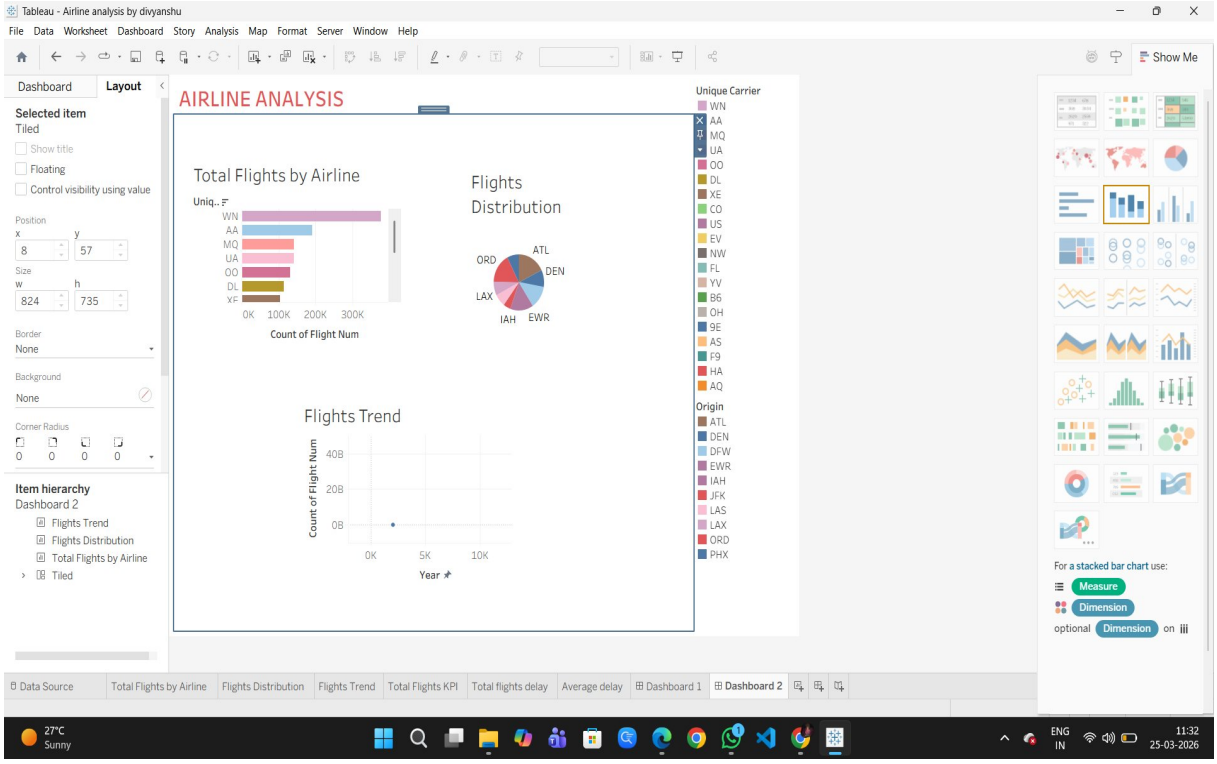


*Figure 1: Dashboard 2 — Airline Analysis with Bar Chart, Pie Chart, and Trend Line*

## Dashboard Description

Dashboard 2 is the main analytical view of the project, consolidating three key visualizations into a single interactive layout. It provides a 360-degree view of airline operations — showing which airlines fly the most, how flights are distributed across major airports, and how flight volume has trended over time.

## Visualizations Included:

### 1. Total Flights by Airline (Horizontal Bar Chart)

This chart ranks all 20 unique airline carriers by total flight count. Carrier WN (Southwest Airlines) leads with the highest count exceeding 300K flights, followed by AA (American Airlines) and MQ. The horizontal layout makes it easy to compare carrier names of varying lengths. Each bar is color-coded by the Unique Carrier dimension for quick visual differentiation.

### 2. Flights Distribution (Pie Chart)

The pie chart shows how flights are distributed across key origin airports including ATL (Atlanta), DEN (Denver), EWR (Newark), IAH (Houston), LAX (Los Angeles), and ORD (Chicago O'Hare). ATL appears to hold the largest share, which aligns with its status as one of the busiest airports in the U.S. This chart is useful for understanding route concentration by origin.

### 3. Flights Trend (Line Chart)

The trend chart plots total flight count (Y-axis) against Year (X-axis), revealing how airline activity has changed over the period covered in the dataset. A single data point is visible, suggesting either a filtered view or a single-year dataset. This chart is useful for monitoring growth or decline in operations over time.

## Filters & Interactivity

Dashboard 2 includes interactive filters for **Unique Carrier** and **Origin Airport**, displayed on the right side. Users can select specific airlines or airports to dynamically update all three charts simultaneously, enabling drill-down exploration of the data.

# Dashboard 1 — KPI Cards

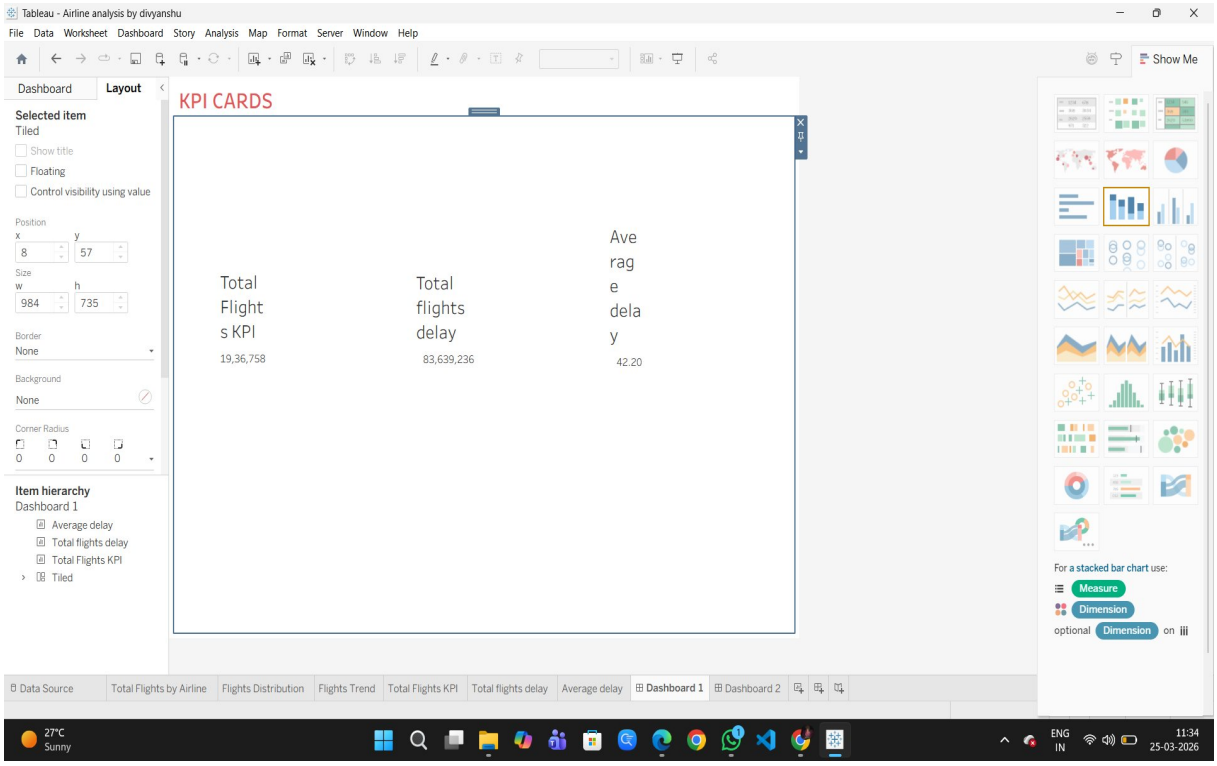


*Figure 2: Dashboard 1 — KPI Cards showing Total Flights, Total Delay, and Average Delay*

## Dashboard Description

Dashboard 1 is a high-level summary view designed to give stakeholders a quick snapshot of the most critical performance metrics. It follows the best practice of keeping dashboards minimal and clear, presenting only the most essential numbers without visual clutter.

## KPI Cards Explained:

### Total Flights KPI — 19,36,758

This KPI card displays the total number of flight records in the dataset. It represents the overall scale of the airline operations being analyzed. Nearly 2 million flights were recorded, indicating a large, multi-year or high-volume dataset.

### Total Flights Delay — 83,639,236

This metric shows the sum of all delay minutes across every flight in the dataset. The enormous total (over 83 million minutes) highlights that delays are a systemic and widespread issue across U.S. airlines. This is a critical operational KPI for airline management.

### Average Delay — 42.20 minutes

The average delay per flight is approximately 42 minutes. This is a key benchmark — any flight delayed beyond this threshold is performing worse than average. It helps operations teams identify outlier routes or carriers with disproportionate delays.

## Design Note

The KPI dashboard uses large, bold typography to make numbers instantly readable. The three metrics are spaced evenly across the dashboard width. This layout is a standard design pattern in business intelligence tools for executive-level reporting.

# Worksheet — Total Flights by Airline

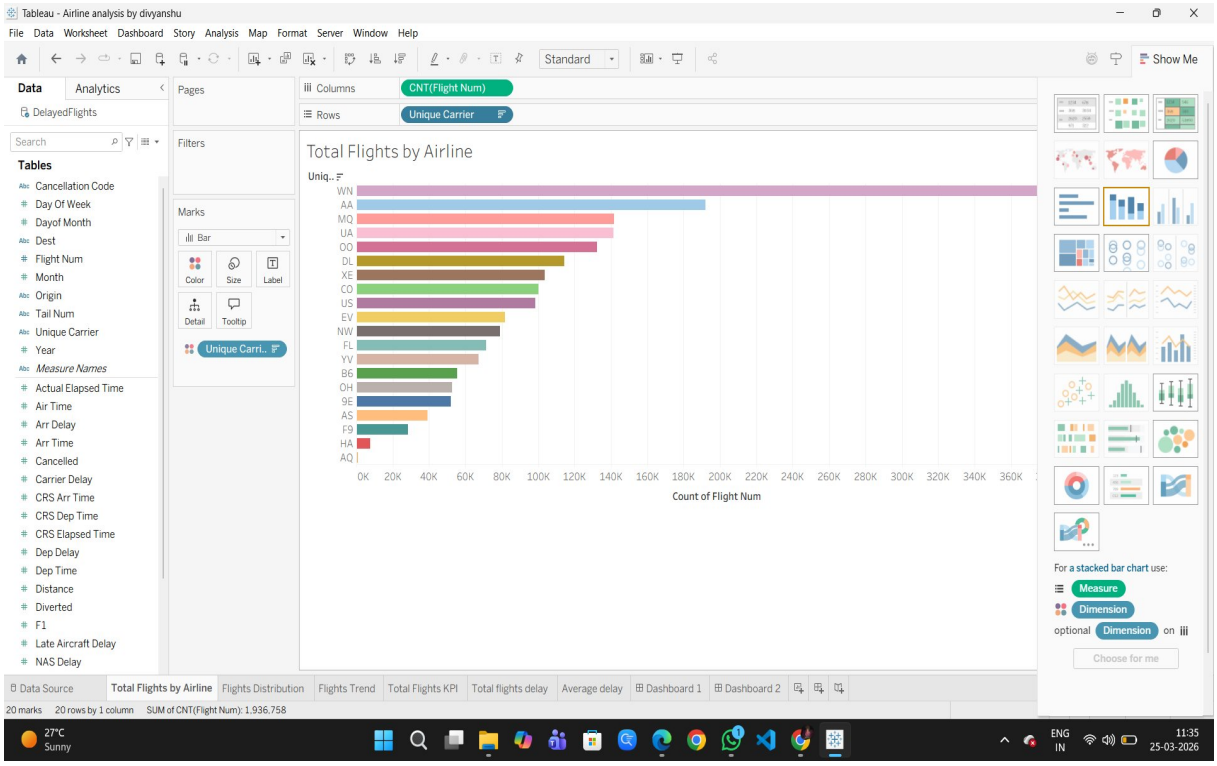


*Figure 3: Detailed Worksheet — Total Flights by Airline (Horizontal Bar Chart)*

## Chart Description

This worksheet provides a detailed, full-screen view of the horizontal bar chart comparing total flights across all 20 airline carriers. The chart is built using **CNT(Flight Num)** on Columns and **Unique Carrier** on Rows, with each carrier color-coded for easy identification.

## Key Observations:

- **WN (Southwest Airlines)** is the dominant carrier with over 360K flights — significantly ahead of all others.
- **AA (American Airlines)** ranks second with approximately 140K flights.
- **MQ and UA** are closely matched in the third and fourth positions.
- **HA and AQ** are at the bottom with minimal flights, indicating smaller regional operations.
- The total sum shown in the status bar is **1,936,758 flights**, confirming the KPI dashboard figure.
- The X-axis scale goes up to 360K, with clear gridlines at 20K intervals for precise reading.

## Data Fields Used:

| Field | Role | Type |
|---|---|---|
| Flight Num | Count (Measure) | X-axis / Value |

| Unique Carrier | Dimension | Y-axis / Category |
| --- | --- | --- |
| Unique Carrier | Color Mark | Visual Differentiation |

# Worksheet — Flights Distribution by Airport

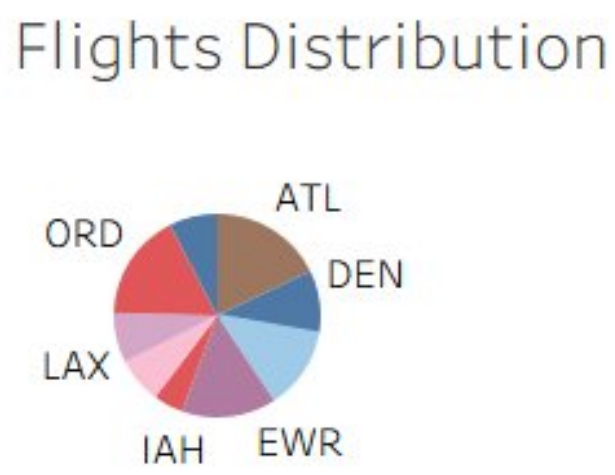


*Figure 4: Flights Distribution — Pie Chart by Origin Airport*

## Chart Description

This pie chart visualizes the proportion of flights departing from each major origin airport. It provides a quick, intuitive understanding of which airports serve as the primary hubs for the airlines in this dataset.

## Airports Represented:

| Code | Airport | Significance |
|---|---|---|
| **ATL** | Atlanta Hartsfield-Jackson | Largest segment, reflecting its status as the world's busiest airport |
| **DEN** | Denver International | Second largest share in the distribution |
| **EWR** | Newark Liberty International | Significant Northeast hub |
| **IAH** | Houston George Bush Intercontinental | Major Southern hub |
| **LAX** | Los Angeles International | Key West Coast gateway |
| **ORD** | Chicago O'Hare International | Major Midwest hub |

## Insight

The fairly even distribution across these six airports suggests that the dataset covers major national hubs rather than being concentrated on a single region. ATL's slightly larger slice is consistent with its position as the highest-traffic airport in the United States. Airlines with heavy ATL presence (like Delta) naturally inflate this segment.

## Summary & Insights

| | |
|---|---|
| **1. Southwest Airlines (WN) dom** | With over 360K flights, WN is the single largest carrier in the dataset — nearly 2.5x t |
| **2. Delays are systemic across th** | An average delay of 42.2 minutes and a total of 83.6 million delay-minutes indicates |
| **3. Atlanta (ATL) is the top origin** | The flights distribution pie chart confirms ATL as the leading origin airport, followed c |
| **4. Dashboard design follows bes** | The use of KPI cards, color-coded bar charts, and interactive filters demonstrates eff |